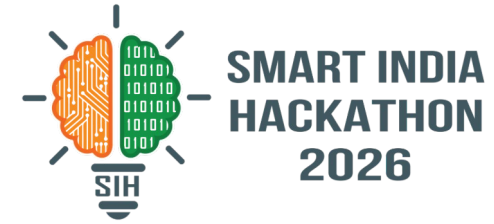
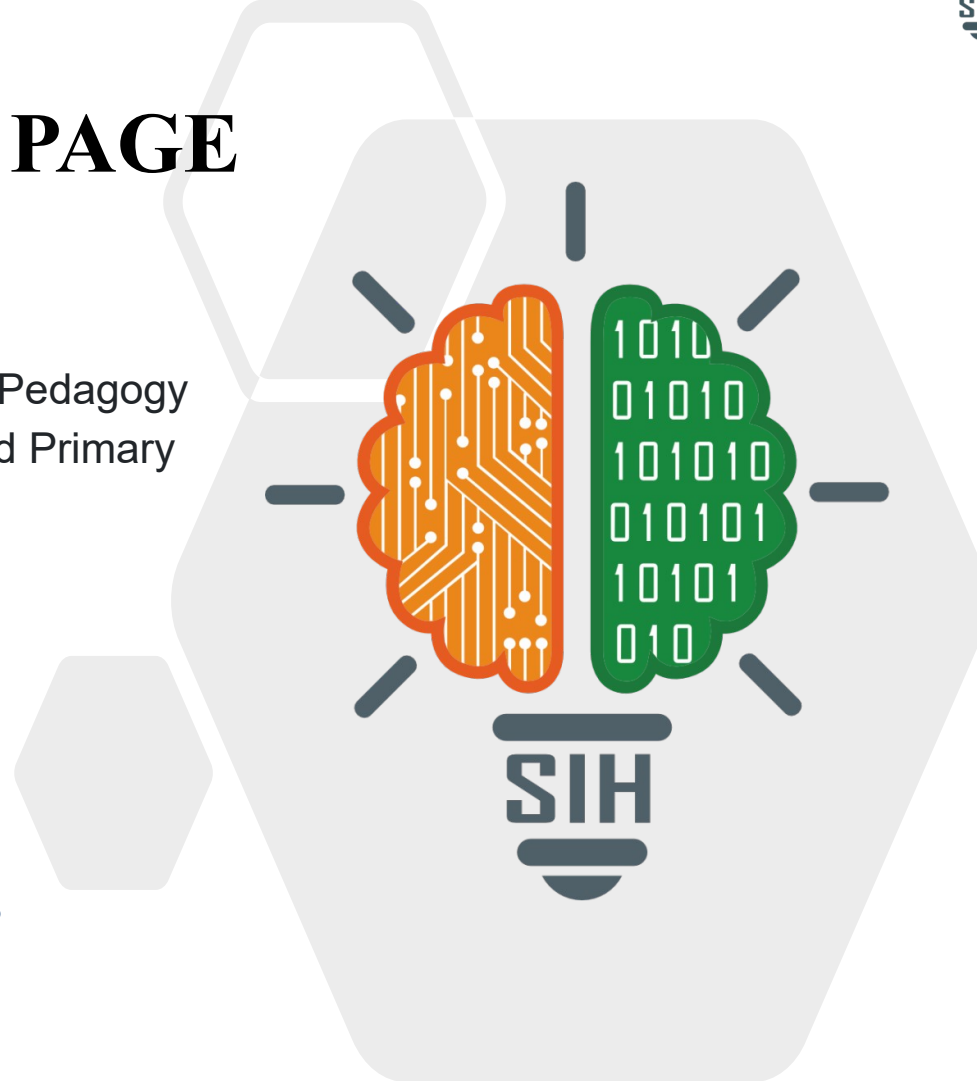


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID** – SIH26042 / 042
- **Problem Statement Title** – AI-Powered Vernacular Pedagogy and Real-Time Translation Tool for Mother Tongue-Based Primary Education
- **Theme** – Education / EdTech
- **PS Category** – Software
- **Team ID** – [ENTER YOUR OFFICIAL SIH TEAM ID]
- **Team Name** (Registered on portal) – Team Bytes



❖ ❖ Proposed Solution (Describe your Idea/Solution/Prototype)

• Detailed Explanation of the Proposed Solution

- **Platform Overview:** AI-powered mobile/web platform translating primary school content into tribal & regional mother tongues in real time.
- **Multimodal Core:** Combines speech-to-text (STT), neural machine translation (NMT), and text-to-speech (TTS) for natural two-way native interaction.
- **Offline Flow:** Generates synchronized bilingual learning materials (text + audio) with offline-first caching for zero-connectivity classrooms.

• How It Addresses the Problem

- **Bridging Curriculum Gap:** Removes language barriers between standardized curricula and mother-tongue learners, fulfilling NEP 2020 primary mandates.
- **Teacher Empowerment:** Provides rural educators with an instant localization tool, eliminating complete reliance on scarce human native translators.

• Innovation and Uniqueness of the Solution

- **Modular Architecture:** Modular language-plugin architecture allows rapid onboarding of new tribal/regional dialects without rebuilding the core engine.
- **Edge-Optimized AI:** Quantized offline on-device AI inference engineered specifically for ultra-low-cost Android devices, unlike cloud-bound tools.

- **Technologies to be Used (Programming Languages, Frameworks, Hardware)**
 - **Client Application:** Flutter / React Native (lightweight, cross-platform, optimized for low-end Android smartphones).
 - **Backend Framework:** Python (FastAPI microservices) for high-efficiency model serving, API orchestration, and sync.
 - **Core AI Models:** IndicTrans2 (Multilingual NMT), OpenAI Whisper & Vosk (offline STT), Indic-TTS & Coqui TTS (native audio).
 - **Edge & Storage:** TensorFlow Lite / ONNX Runtime for INT8 quantized offline execution; SQLite (offline cache) & PostgreSQL.
- **Methodology and Process for Implementation (Flow Charts / Images / Working Prototype)**
 - **End-to-End Pipeline:** Step 1 (Input Content) → Step 2 (Speech-to-Text) → Step 3 (Neural Translation) → Step 4 (TTS Audio Generation) → Step 5 (Bilingual Delivery & Offline Caching).



- **Analysis of the Feasibility of the Idea**

- **Proven AI Foundations:** Built on proven AI/ML models (IndicTrans2, Whisper, Indic-TTS) with verified benchmarks across Indian languages.
- **Hardware Compatibility:** Engineered specifically for low-end Android smartphones prevalent in rural and tribal schooling environments.
- **Modular Scalability:** Modular language-plugin architecture allows rapid expansion to new Indian and tribal dialects without system refactoring.
- **Zero-Cost Offline Flow:** Offline-first functionality eliminates ongoing internet data costs and operates reliably in remote shadow regions.

- **Potential Challenges and Risks**

- **Dataset Scarcity:** Limited parallel datasets and acoustic corpora available for low-resource tribal languages (e.g. Santhali, Mundari, Ho).
- **Linguistic Diversity:** Dialectical variations and pronunciation nuances affecting real-time speech recognition and translation accuracy.
- **Hardware Latency:** Real-time inference latency and RAM/storage constraints on entry-level Android devices.

- **Strategies for Overcoming These Challenges**

- **Quantized Edge Runtime:** Deploy INT8 quantized TFLite / ONNX models to achieve sub-second on-device inference with minimal memory footprint.
- **Intelligent Local Cache:** Locally pre-cache frequently used primary curriculum units and lessons for instant zero-latency classroom playback.
- **Community Data Expansion:** Continuously expand tribal language resources via validated educational corpora and community-in-the-loop feedback.

- **Potential Impact on the Target Audience**
 - **NEP 2020 Alignment:** Directly operationalizes the National Education Policy (NEP 2020) mandate for mother-tongue foundational schooling.
 - **Inclusive Access:** Prevents early childhood cognitive exclusion and learning dropouts among tribal and vernacular-speaking children.
 - **Enhanced Learning Gains:** Bridges the comprehension gap between state-level textbooks and local dialects, accelerating primary literacy and numeracy.
- **Benefits of the Solution (Social, Economic, Environmental, etc.)**
 - **Teacher Assistance:** Empowers rural primary teachers to instantly generate interactive bilingual text and audio learning materials.
 - **Voice-Based Inclusivity:** Voice-first multimodal capability enables non-literate and early-grade learners to participate and comprehend intuitively.
 - **Classroom Engagement:** Eliminates language friction between teachers and pupils, fostering interactive, confident classroom environments.
 - **Linguistic Scalability:** Readily scalable across additional regional and indigenous dialects without costly infrastructural overhaul.
 - **Rural Resilience:** 100% offline capability ensures uninterrupted, equitable learning in remote rural tribal schools with zero connectivity.

- **Details / Links of the Reference and Research Work**

- **Problem Statement:** Government of Jharkhand — Smart India Hackathon 2026 Problem Statement ID: SIH26042 / 042.
- **SIH 2026 Platform:** Official Smart India Hackathon 2026 Portal (<https://sih.gov.in>) — Guidelines & EdTech Domain Specifications.
- **Neural Translation:** AI4Bharat IndicTrans2: Towards High-Quality & Accessible Machine Translation for Indian Languages (Gala et al., 2023).
- **Speech-to-Text Models:** OpenAI Whisper (Robust Speech Recognition via Weak Supervision) & AlphaCephei Vosk Offline Embedded STT.
- **Text-to-Speech Engine:** Indic-TTS (Multilingual TTS for Indian Languages, IIT Madras) & Coqui TTS Deep-Learning Audio Synthesis.
- **Tribal Datasets:** Bhashini (National Language Translation Mission, MeitY) & AI4Bharat Open Linguistic Corpora for Indian & Tribal Languages.
- **Policy Guidelines:** National Education Policy (NEP 2020), Ministry of Education, Govt. of India — Multilingual & Mother-Tongue Pedagogy.